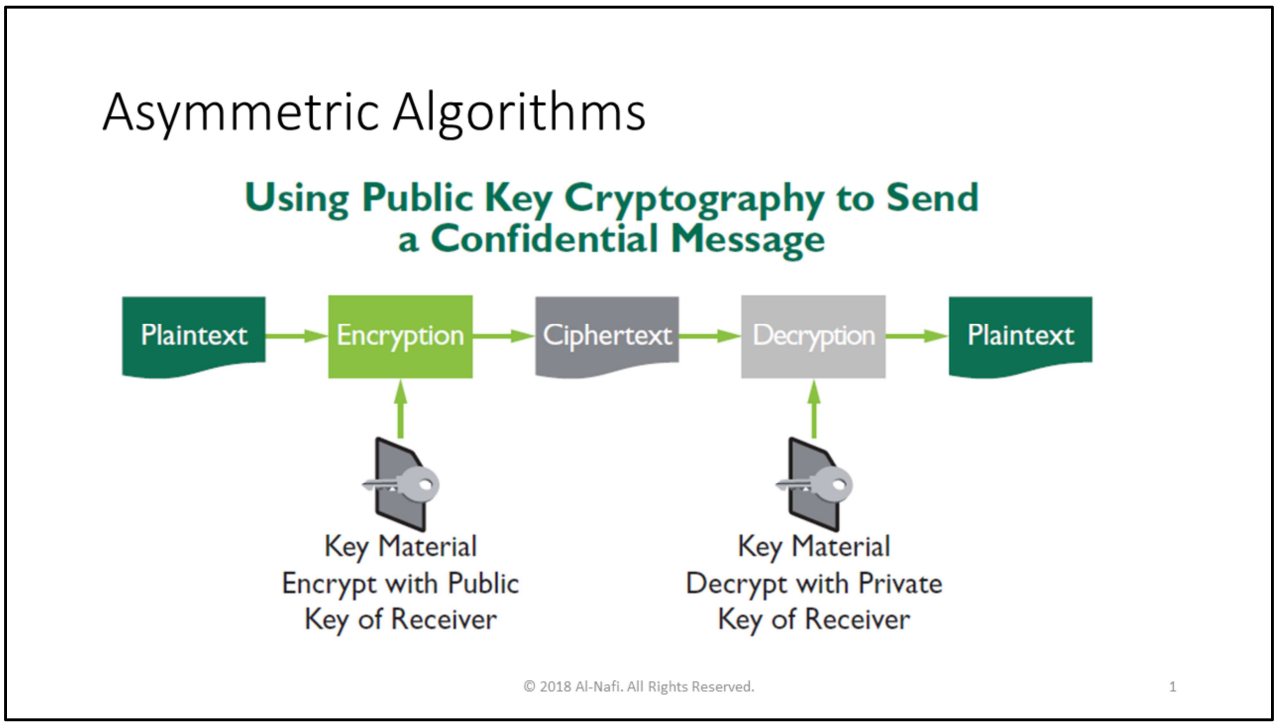


Reference https://searchsecurity.techtarget.com/definition/asymmetric-cryptography

Asymmetric cryptography, also known as public key cryptography, uses public and private keys to encrypt and decrypt data. The keys are simply large numbers that have been paired together but are not identical (asymmetric). One key in the pair can be shared with everyone; it is called the public key. The other key in the pair is kept secret; it is called the private key. Either of the keys can be used to encrypt a message; the opposite key from the one used to encrypt the message is used for decryption.

Many protocols like SSH, OpenPGP, S/MIME, and SSL/TLS rely on asymmetric cryptography for encryption and digital signature functions. It is also used in software programs, such as browsers, which need to establish a secure connection over an insecure network like the Internet or need to validate a digital signature. Encryption strength is directly tied to key size and doubling the key length delivers an exponential increase in strength, although it does impair performance. As computing power increases and more efficient factoring algorithms are discovered, the ability to factor larger and larger numbers also increases.

For asymmetric encryption to deliver confidentiality, integrity, authenticity and non-repudiability, users and systems need to be certain that a public key is authentic, that it belongs to the person or entity claimed and that it has not been tampered with nor replaced by a malicious third party. There is no perfect solution to this public key authentication problem. A public key infrastructure (PKI) -- where trusted certificate authorities certify ownership of key pairs and certificates -- is the most common approach, but encryption products based on the Pretty Good Privacy (PGP) model -- including OpenPGP -- rely on a decentralized authentication model called a web of trust, which relies on individual endorsements of the link between user and public key.

How asymmetric encryption works

Asymmetric encryption algorithms use a mathematically-related key pair for encryption and decryption; one is the public key and the other is the private key. If the public key is used for encryption, the related private key is used for decryption and if the private key is used for encryption, the related public key is used for decryption.

Only the user or computer that generates the key pair has the private key. The public key can be distributed to anyone who wants to send encrypted data to the holder of the private key. It's impossible to determine the private key with the public one.

The two participants in the asymmetric encryption workflow are the sender and the receiver. First, the sender obtains the receiver's public key. Then the plaintext is encrypted with the asymmetric encryption algorithm using the recipient's public key, creating the ciphertext. The ciphertext is then sent to the receiver, who decrypts the ciphertext with his private key so he can access the sender's plaintext.

Because of the one-way nature of the encryption function, one sender is unable to read the messages of another sender, even though each has the public key of the receiver.

Examples of asymmetric cryptography

RSA (Rivest-Shamir-Adleman) -- the most widely used asymmetric algorithm -- is embedded in the SSL/TSL protocols which is used to provide communications security over a computer network. RSA derives its security from the computational difficulty of factoring large integers that are the product of two large prime numbers. Multiplying two large primes is easy, but the difficulty of determining the original numbers from the product -- factoring -- forms the basis of public key cryptography security. The time it takes to factor the product of two sufficiently large primes is considered to be beyond the capabilities of most attackers, excluding nation-state actors who may have access to sufficient computing power. RSA keys are typically 1024- or 2048-bits long, but experts believe that 1024-bit keys could be broken in the near future, which is why government and industry are moving to a minimum key length of 2048-bits.

Elliptic Curve Cryptography (ECC) is gaining favor with many security experts as an

alternative to RSA for implementing public key cryptography. ECC is a public key encryption technique based on elliptic curve theory that can create faster, smaller, and more efficient cryptographic keys. ECC generates keys through the properties of the elliptic curve equation.

To break ECC, one must compute an elliptic curve discrete logarithm, and it turns out that this is a significantly more difficult problem than factoring. As a result, ECC key sizes can be significantly smaller than those required by RSA yet deliver equivalent security with lower computing power and battery resource usage making it more suitable for mobile applications than RSA.

Uses of asymmetric cryptography

The typical application for asymmetric cryptography is authenticating data through the use of digital signatures. Based on asymmetric cryptography, digital signatures can provide assurances of evidence to the origin, identity and status of an electronic document, transaction or message, as well as acknowledging informed consent by the signer.

To create a digital signature, signing software -- such as an email program -- creates a one-way hash of the electronic data to be signed. The user's private key is then used to encrypt the hash, returning a value that is unique to the hashed data. The encrypted hash, along with other information such as the hashing algorithm, forms the digital signature. Any change in the data, even to a single bit, results in a different hash value.

This attribute enables others to validate the integrity of the data by using the signer's public key to decrypt the hash. If the decrypted hash matches a second computed hash of the same data, it proves that the data hasn't changed since it was signed. If the two hashes don't match, the data has either been tampered with in some way -- indicating a failure of integrity -- or the signature was created with a private key that doesn't correspond to the public key presented by the signer -- indicating a failure of authentication.

A digital signature also makes it difficult for the signing party to deny having signed something -- the property of non-repudiation. If a signing party denies a valid digital signature, their private key has either been compromised or they are being untruthful. In many countries, including the United States, digital signatures have the same legal weight as more traditional forms of signatures.

Asymmetric cryptography can be applied to systems in which many users may need to encrypt and decrypt messages, such as encrypted email, in which a public key can be used to encrypt a message, and a private key can be used to decrypt it.

The SSL/TSL cryptographic protocols for establishing encrypted links between websites and browsers also make use of asymmetric encryption.

Additionally, Bitcoin and other cryptocurrencies rely on asymmetric cryptography as users have public keys that everyone can see and private keys that are kept secret. Bitcoin uses a cryptographic algorithm to ensure that only the legitimate

owners can spend the funds.
In the case of the Bitcoin ledger, each unspent transaction output (UTXO) is typically associated with a public key. So if user X, who has an UTXO associated with his public key, wants to send the money to user Y, user X uses his private key to sign a transaction that spends the UTXO and creates a new UTXO that's associated with user Y's public key.

Asymmetric vs. symmetric cryptography

The main difference between these two methods of encryption is that asymmetric encryption algorithms makes use of two different but related keys -- one key to encrypt the data and another key to decrypt it -- while symmetric encryption uses the same key to perform both the encryption and decryption functions.
Another difference between asymmetric and symmetric encryption is the length of the keys. In symmetric cryptography, the length of the keys -- which is randomly selected -- are typically set at 128-bits or 256-bits, depending on the level of security that's needed.
However, in asymmetric encryption, there has to be a mathematical relationship between the public and private keys. Because hackers can potentially exploit this pattern to crack the encryption, asymmetric keys need to be much longer to offer the same level of security. The difference in the length of the keys is so pronounced that a 2048-bit asymmetric key and a 128-bit symmetric key provide just about an equivalent level of security.
Additionally, asymmetric encryption is slower than symmetric encryption, which has a faster execution speed.

History of asymmetric cryptography

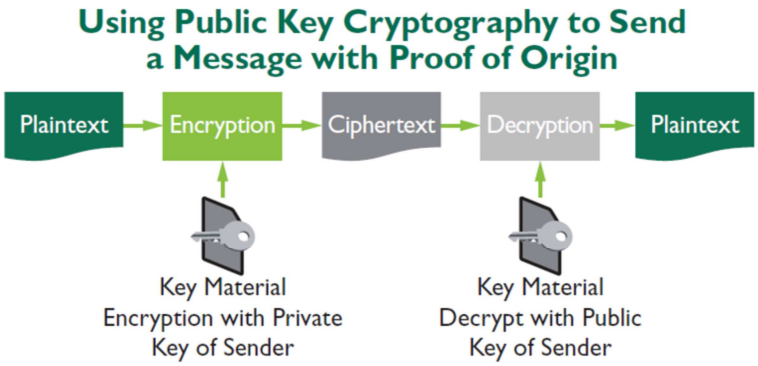
Whitfield Diffie and Martin Hellman, researchers at Stanford University, first publicly proposed asymmetric encryption in their 1977 paper, "New Directions in Cryptography." The concept had been independently and covertly proposed by James Ellis several years earlier, while he was working for the Government Communications Headquarters (GCHQ), the British intelligence and security organization. The asymmetric algorithm as outlined in the Diffie-Hellman paper uses numbers raised to specific powers to produce decryption keys. Diffie and Hellman had initially teamed up in 1974 to work on solving the problem of key distribution problem.
The RSA algorithm, which was based on the work of Diffie, was named after its three inventors -- Ronald Rivest, Adi Shamir and Leonard Adleman. They invented the RSA algorithm in 1977, and published it in *Communications of the ACM* in 1978.
Today, RSA is the standard asymmetric encryption algorithm and it's used in many areas, including TLS/SSL, SSH, digital signatures and PGP.

Benefits and disadvantages of asymmetric cryptography

The benefits of asymmetric cryptography include:
the key distribution problem is eliminated because there's no need for exchanging keys.

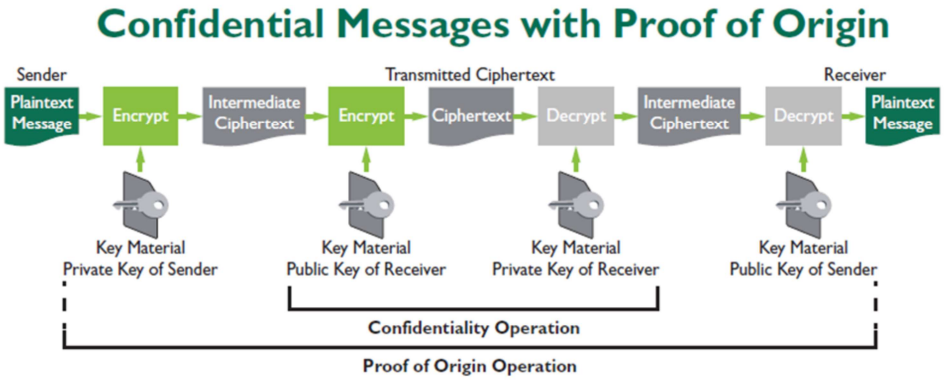
security is increased as the private keys don't ever have to be transmitted or revealed to anyone.
the use of digital signatures is enabled so that a recipient can verify that a message comes from a particular sender.
it allows for non-repudiation so the sender can't deny sending a message.
Disadvantages include:
it's a slow process compared to symmetric crytography, so it's not appropriate for decrypting bulk messages.
if an individual loses his private key, he can't decrypt the messages he receives.
since the public keys aren't authenticated, no one really knows if a public key belongs to the person specified. Consequently, users have to verify that their public keys belong to them.
if a hacker identifies a person's private key, the attacker can read all of that individual's messages.

Asymmetric Algorithms



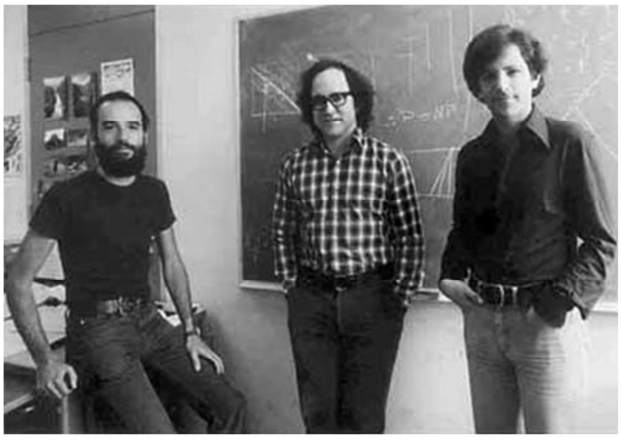
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Asymmetric Algorithms



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Rivest-Shamir-Adleman (RSA) Algorithm



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Reference

The RSA algorithm is the basis of a cryptosystem -- a suite of cryptographic algorithms that are used for specific security services or purposes -- which enables public key encryption and is widely used to secure sensitive data, particularly when it is being sent over an insecure network such as the internet.

RSA was first publicly described in 1977 by Ron Rivest, Adi Shamir and Leonard Adleman of the Massachusetts Institute of Technology, though the 1973 creation of a public key algorithm by British mathematician Clifford Cocks was kept classified by the U.K.'s GCHQ until 1997.

Public key cryptography, also known as asymmetric cryptography, uses two different but mathematically linked keys -- one public and one private. The public key can be shared with everyone, whereas the private key must be kept secret.

In RSA cryptography, both the public and the private keys can encrypt a message; the opposite key from the one used to encrypt a message is used to decrypt it. This attribute is one reason why RSA has become the most widely used asymmetric algorithm: It provides a method to assure the confidentiality, integrity, authenticity, and non-repudiation of electronic communications and data storage.

Many protocols like Secure Shell, OpenPGP, S/MIME, and SSL/TLS rely on RSA for encryption and digital signature functions. It is also used in software programs -- browsers are an obvious example, as they need to establish a secure connection over an insecure network, like the internet, or validate a digital signature. RSA signature verification is one of the most commonly performed operations in network-connected systems.

Why the RSA algorithm is used

RSA derives its security from the difficulty of factoring large integers that are the product of two large prime numbers. Multiplying these two numbers is easy, but determining the original prime numbers from the total -- or factoring -- is considered infeasible due to the time it would take using even today's supercomputers.

The public and private key generation algorithm is the most complex part of RSA cryptography. Two large prime numbers, p and q, are generated using the Rabin-Miller primality test algorithm. A modulus, n, is calculated by multiplying p and q. This number is used by both the public and private keys and provides the link between them. Its length, usually expressed in bits, is called the key length.

Alan Way explains how RSA public key encryption works

The public key consists of the modulus n and a public exponent, e, which is normally set at 65537, as it's a prime number that is not too large. The e figure doesn't have to be a secretly selected prime number, as the public key is shared with everyone.

The private key consists of the modulus n and the private exponent d, which is calculated using the Extended Euclidean algorithm to find the multiplicative inverse with respect to the totient of n.

Read on or watch the video below for a more detailed explanation of how the RSA algorithm works.

How does the RSA algorithm work?

Alice generates her RSA keys by selecting two primes: p=11 and q=13. The modulus is n=p×q=143. The totient is n φ(n)=(p−1)x(q−1)=120. She chooses 7 for her RSA public key e and calculates her RSA private key using the Extended Euclidean algorithm, which gives her 103.

Bob wants to send Alice an encrypted message, M, so he obtains her RSA public key (n, e) which, in this example, is (143, 7). His plaintext message is just the number 9 and is encrypted into ciphertext, C, as follows:

$M^e \bmod n = 9^7 \bmod 143 = 48 = C$

When Alice receives Bob's message, she decrypts it by using her RSA private key (d, n) as follows:

$C^d \bmod n = 48^{103} \bmod 143 = 9 = M$

To use RSA keys to digitally sign a message, Alice would need to create a hash -- a message digest of her message to Bob -- encrypt the hash value with her RSA private

key, and add the key to the message. Bob can then verify that the message has been sent by Alice and has not been altered by decrypting the hash value with her public key. If this value matches the hash of the original message, then only Alice could have sent it -- authentication and non-repudiation -- and the message is exactly as she wrote it -- integrity.
Alice could, of course, encrypt her message with Bob's RSA public key -- confidentiality -- before sending it to Bob. A digital certificate contains information that identifies the certificate's owner and also contains the owner's public key. Certificates are signed by the certificate authority that issues them, and they can simplify the process of obtaining public keys and verifying the owner.

RSA security
RSA security relies on the computational difficulty of factoring large integers. As computing power increases and more efficient factoring algorithms are discovered, the ability to factor larger and larger numbers also increases.

Encryption strength is directly tied to key size, and doubling key length can deliver an exponential increase in strength, although it does impair performance. RSA keys are typically 1024- or 2048-bits long, but experts believe that 1024-bit keys are no longer fully secure against all attacks. This is why the government and some industries are moving to a minimum key length of 2048-bits.
Barring an unforeseen breakthrough in quantum computing, it will be many years before longer keys are required, but elliptic curve cryptography (ECC) is gaining favor with many security experts as an alternative to RSA to implement public key cryptography. It can create faster, smaller and more efficient cryptographic keys. Modern hardware and software are ECC-ready, and its popularity is likely to grow, as it can deliver equivalent security with lower computing power and battery resource usage, making it more suitable for mobile apps than RSA. Finally, a team of researchers, which included Adi Shamir, a co-inventor of RSA, has successfully created a 4096-bit RSA key using acoustic cryptanalysis; however, any encryption algorithm is vulnerable to attack.

RSA: Rivest Shamir Adleman

- **RSA** is one of the first practical public-key algorithm and is widely used for secure data transmission.
- Of all the public-key algorithms proposed over the years, RSA is by far the easiest to understand and implement.
- The algorithm is named after the three inventors—Ron Rivest, Adi Shamir, and Leonard Adleman
- The RSA algorithm was publicly described in 1977 by Rivest, Shamir, and Adleman at MIT

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